

Swapnil Tukaram Mane

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SUMMARY

Software engineer with 5+ years building AI and full-stack products. Shipped production machine learning and LLM systems across enterprise and startup environments, working in Python, Java, and Rust on AWS, GCP, and Azure. Led an engineering team and established its code review and CI/CD practices.

TECHNICAL SKILLS

AI & LLM Systems: LLM inference, RAG, AI agents, multi-agent systems (AutoGen), agentic workflows, MCP, prompt engineering, prompt-injection defense, evals, model deployment

Languages: Python, Java, TypeScript, Rust, Kotlin, C#/.NET, SQL

Backend & Full-Stack: Node.js, Express, React, Next.js, REST APIs, distributed systems, PostgreSQL, SQL Server, Redis, Playwright

Data & ML: PySpark, Spark tuning, ETL pipelines, feature engineering, imbalanced data (SMOTE), TF-IDF, NLP, LSTM forecasting, model evaluation, Azure Data Lake, Tableau

Cloud & Practices: AWS, GCP, Azure Machine Learning, GPU inference, cloud cost optimization, CI/CD, GitHub Actions

PROFESSIONAL EXPERIENCE

Asearis Data Systems Inc., Lead Software Engineer

July 2025 - Present

- Owned the Java and PostgreSQL backend of a distributed LLM inference platform, including chat-history persistence, tokenization, and the per-user and per-model usage metering that underpinned customer billing
- Integrated on-device AI inference into the Android and C#/.NET Windows applications, resolving the memory constraints that had caused crashes on mobile hardware
- Led sprint planning across Java, Rust, and mobile deliverables, established PR-based code review and CI/CD as team standards, and delivered the authentication and account-security hardening required for SOC 2 readiness

Sitewiz, Inc., AI Engineer

Aug 2025 - Sep 2025

- Led a technical review of the cloud architecture that identified redundant GPU capacity, then consolidated inference onto dedicated AWS endpoints, reducing spend by approximately \$16,000 per month and 5+ engineering hours per week
- Resolved the state-management failures affecting the AutoGen multi-agent workflows, and hardened a set of experimental features into production services ahead of an enterprise launch

Bridge Green Upcycle Corp, Software Engineering Intern

Dec 2023 – Dec 2024

- Built a PySpark data pipeline over 18M+ battery records that reduced processing time by 95%. Established the AI/ML team's architecture, led product development, and onboarded new engineers
- Trained an LSTM-DNN model forecasting battery health to 5.7% MAE and a classification model grading batteries for reuse at approximately 90% accuracy, delivered through Python and Tableau dashboards used for the Series A raise
- Applied physics-based feature extraction, including capacity estimation and open-circuit voltage profiling, for real-time battery state-of-health estimation that enabled predictive maintenance scheduling and failure analysis

LTIMindtree Ltd. (formerly Larsen & Toubro Infotech Ltd.), Software Engineer

Aug 2020 – Jul 2023

- Trained an NLP classification model on 900K+ support tickets in Azure ML and deployed it as a live inference endpoint within the ticketing application, automating ticket routing and reducing time to resolution
- Received the GoMx Individual Excellence Award for two automation tools: a .NET utility that reduced a 35-minute system check to 5 minutes, and a Python OCR search tool that reduced lookup time by 67%
- Delivered SAP ABAP development on S/4HANA for enterprise clients, working directly with the client and their functional consultants across MM, SD, and FI, and authored the training and handover documentation their teams operated from

PROJECTS

Portfolio RAG Chatbot (Demo)

Apr 2026

- Built a retrieval-based assistant that answered strictly from a verified knowledge source, with a two-stage intent router that screened off-topic and prompt-injection attempts before the model call, reducing cost and protecting answer accuracy. Reached 95.8% precision, 95.4% F1, and approximately 3.2s response time

Live Co-pilot, Real-Time Meeting Assistant (Demo)

Apr 2026

- Built a real-time meeting assistant on Next.js and React with Groq Whisper and GPT-OSS 120B that generated contextual suggestions during a call, each linked to the exact transcript line it came from, supported by an automated evaluation suite that blocked low-quality output

Multi-Agent Text-to-SQL (GitHub)

Jun 2025

- Built a multi-agent system on AutoGen and Ollama that translated natural-language questions into executable SQL, giving non-technical users direct access to the database. Achieved 60% execution accuracy on the Bird.dev benchmark

EDUCATION

Binghamton University (SUNY), Master of Science in Computer Science, GPA 3.71/4.0

Aug 2023 - May 2025